MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL SEMESTER 2025
CHAPTER 2: SEQUENCES (PART 3)

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Goal of this section:

(1) Series
(2) Convergence Tests
(3) Absolute convergence

Reference books:

- Robert G. Bartle, Donald R. Sherbert: Introduction to Real Analysis (Wiley, 2011, Fourth Edition)
- Kenneth A. Ross: Elementary Analysis: The Theory of Calculus (Springer, 2013, Second Edition)

1. Series

1.1. **Definition.**

Definition 1.1. Let (x_n) be a sequence in $\mathbb{R}$. Then we define the series to be:

$$\begin{aligned} s_1 &= x_1 \\ s_2 &= x_1 + x_2 \\ s_3 &= x_1 + x_2 + x_3 \\ &\vdots \\ s_k &= x_1 + \ldots + x_k \end{aligned}$$

One may regard (s_k) to be a sequence, and so it makes sense to write $\lim\ s_k$. We say that the series s_k is convergent if $\lim\ s_k$ exists. Otherwise, we say that the series s_k is divergent.

We shall say the terms $\sum_{k=1}^{n} x_k$ to be partial sums. Sometimes partial sums can also start from some latter terms, such as $\sum_{k=5}^{n} x_k$.

1.2. **Examples.**

Example 1.2. (Series) For $r > 0$, $\sum r^n$.

(i) If $r < 1$, then $\sum_{k=1}^{\infty} r^k$ converges and $\sum r^k = \frac{r}{1-r}$. This can be shown by the formula $\sum_{k=1}^{n} r^n = \frac{r-r^{n+1}}{1-r}$.
(ii) If $r \geq 1$, then $\sum r^n$ diverges.

Example 1.3. (p-harmonic series) Let $x_n = \frac{1}{n^p}$. Let

$$s_n = 1 + \frac{1}{2^p} + \ldots + \frac{1}{n^p}.$$

We shall consider $p = 2$. Then,

$$s_1 = 1, \quad s_2 = 1 + \frac{1}{2^2} < 1 + \frac{1}{2}$$

$$s_4 = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} < 1 + \frac{1}{2^2} + \frac{1}{2^2} + \frac{1}{4^2} < 1 + \frac{1}{2} + \frac{1}{4}$$

$$s_{2^k} < 1 + \frac{1}{2} + \ldots + \frac{1}{2^k} < 2.$$

Thus, since (s_n) is increasing and (s_{2^k}) is bounded, (s_n) is also bounded. Since (s_n) is also monotone, the series converges.

Exercise 1.4. For $p \geq 2$, show that the p-harmonic series converges.

Example 1.5. Let $x_n = \frac{1}{n}$. Let

$$s_n = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$$

Note that the series (s_n) is monotone. However, we shall show that it is not bounded.

We shall consider the terms s_{2^k-1}:

(1) $s_1 = 1$
(2) $s_3 = 1 + \frac{1}{2} + \frac{1}{3} > 1 + \frac{1}{4} + \frac{1}{4} = 1 + \frac{1}{2}$
(3) $s_7 = 1 + \left(\frac{1}{2} + \frac{1}{3}\right) + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} > 1 + \frac{1}{2} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = 1 + \frac{1}{2} + \frac{1}{2}$.

Note that, for $k \geq 3$,

$$s_{2^{k+1}-1} - s_{2^k-1} = \frac{1}{2^k} + \cdots \frac{1}{2^{k+1}-1} > \frac{1}{2^{k+1}} + \cdots \frac{1}{2^{k+1}} = \frac{1}{2}.$$

Thus, $s_{2^k-1} > 1 + \frac{k-1}{2}$, and so the sequence (s_n) is not bounded. Hence, (s_n) is not convergent.

2. Some Convergence Tests

2.1. Convergence Tests.

Theorem 2.1. *(n-th series test) If the series $\sum x_n$ converges, then* $\lim\ x_n = 0$.

Proof. Let $s_n = \sum_{k=1}^{n} x_k$. Since the series converges, $\lim\ s_n$ exists. Now $x_n = s_n - s_{n-1}$, and so $\lim\ x_n = \lim\ (s_n - s_{n-1}) = \lim\ s_n - \lim\ s_{n-1} = 0$. □

The convergence test is useful for showing a series to be *divergent*.

Example 2.2. Let $x_n = \frac{n}{n+1}$. Then $\sum \frac{n}{n+1}$ diverges since $\lim \frac{n}{n+1} = 1 \neq 0$.

Exercise 2.3. Let $x_n = (-1)^n n$. Determine whether the series $\sum_n x_n$ converges.

Exercise 2.4. Discuss whether the converse of Theorem 2.1 holds.

We have the following criteria from MCT.

Theorem 2.5. *Let (x_n) be a sequence of non-negative real numbers. Then $\sum x_n$ converges if and only if the sequence of the partial sums is bounded.*

Proof. Let

$$s_n = \sum_{k=1}^{n} x_k.$$

Since x_n are non-negative, the sequence (s_n) of partial sums is monotone. Then the theorem follows from the Monotone Convergence Theorem. □

Theorem 2.6. *(Comparison Tests) Let (x_n) and (y_n) be two sequences in $\mathbb{R}$. Suppose for any $n \in \mathbb{N}$,*

$$0 \leq x_n \leq y_n.$$

Then

1. *if $\sum\ y_n$ converges, then $\sum\ x_n$ converges.*
2. *if $\sum\ x_n$ diverges, then $\sum\ y_n$ diverges.*

Proof. Since the partial sums $\sum_{k=1}^{n} y_k$ converge and so is bounded, $\sum_{k=1}^{n} x_k$ is also bounded. Then, the series $\sum_{k=1}^{\infty} x_k$ is also bounded. This shows (a) by Theorem 2.5. (b) is the negation of the statement (a). □

Exercise 2.7.

1. Let $x_1 = 1$ and $x_n = \frac{1}{n(n-1)}$. Use the equation $\frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$ to show that the sequence of partial sums $\sum_{k=1}^{n} x_k$ is bounded above.
2. Use this and Theorem 2.6 to conclude the 2-harmonic series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

Example 2.8. The series $\sum \frac{1}{n^2+2n}$ converges.
Proof: For any $n \in \mathbb{N}$,

$$0 < \frac{1}{n^2+2n} < \frac{1}{n^2}$$

Since $\sum \frac{1}{n^2}$ converges, we have that $\sum \frac{1}{n^2+2n}$ also converges.

Theorem 2.9. *(Limit comparison test) Let (x_n) and (y_n) be sequences of strictly positive numbers. Suppose*

$$r := \lim \left(\frac{x_n}{y_n}\right)$$

exists.

(a) *If $r \neq 0$, then $\sum x_n$ is convergent if and only if $\sum y_n$ is convergent.*
(b) *If $r = 0$ and if $\sum y_n$ is convergent, then $\sum x_n$ is convergent.*

Proof. We first prove (a). One can find an integer $N \in \mathbb{N}$ such that for any $n > N$ (how?),

$$\frac{1}{2}ry_n < x_n < \frac{3}{2}ry_n.$$

Now using $x_n < 2ry_n$ and Theorem 2.6, one deduces the if statement. Similarly, using $\frac{1}{2}ry_n < x_n$ and Theorem 2.6, one deduces the only if statement.

We now prove (b). Let $\epsilon = 1$. By the definition of limit, there exists an integer $N \in \mathbb{N}$ such that for any $n > N$,

$$0 < \frac{x_n}{y_n} < \epsilon = 1$$

and so $x_n \leq y_n$. Now (b) follows from Theorem 2.6. □

Example 2.10. The series $\sum \frac{1}{n^3-n^2+1}$ converges.
Proof: Note that

$$\lim \frac{\frac{1}{n^3-n+1}}{\frac{1}{n^3}} = \lim \frac{n^3}{n^3-n+1} = \lim \frac{1}{1-\frac{1}{n}+\frac{1}{n^3}} = 1.$$

Since $\sum \frac{1}{n^3}$ converges, by Theorem 2.9, $\sum_{n=1}^{\infty} \frac{1}{n^3-n+1}$ converges.

3. Absolute and conditional Convergent series

3.1. **Absolute convergence.**

Definition 3.1. A series $\sum x_k$ in $\mathbb{R}$ is said to be absolutely convergent if $\sum |x_k|$ is convergent in $\mathbb{R}$.

Example 3.2. The series $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}$ is absolutely convergent because 2-harmonic series converges.

Theorem 3.3. *(Absolute convergence ⇒ convergence) Let $\sum_{k=1}^{\infty} x_k$ be an absolutely convergent series in $\mathbb{R}$. Then the series $\sum_{k=1}^{\infty} x_k$ is also convergent.*

Proof. For all $n \in \mathbb{N}$,

$$0 \leq x_n + |x_n| \leq 2|x_n|.$$

Since $\sum\ 2|x_k|$ converges, by Theorem 2.6, the series $\sum_{k=1}^{\infty}(x_k + |x_k|)$ also converges. Thus, $\sum_{k=1}^{\infty} x_k$ converges.

□

Example 3.4. The series $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}$ is absolutely convergent and so $\sum_{n=1}^{\infty} \frac{(-1)^k}{k^2}$ also converges.

Exercise 3.5. Determine if the following series are convergent:

(a) $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2+1}$; and
(b) $\sum_{k=1}^{\infty} \frac{(-1)^k \sin(k)}{k^2}$.

3.2. **Rearrangement Theorem.**

Definition 3.6. A series $\sum y_n$ in $\mathbb{R}$ is said to be a rearrangement of a series $\sum\ x_n$ if there is a bijection $f : \mathbb{N} \to \mathbb{N}$ such that $y_n = x_{f(n)}$ for all $n \in \mathbb{N}$.

Theorem 3.7. *(Rearrangement theorem) Let $\sum\ x_n$ be an absolutely convergent series. Suppose*

$$\sum x_n$$

converges to L. Then any rearrangement $\sum y_n$ of $\sum\ x_n$ also converges to L.

Proof. (Sketch) Let

$$t_n = \sum_{k=1}^{n} y_k, \quad s_n = \sum_{k=1}^{n} x_n.$$

Let f be the function defining the rearrangement. For sufficiently large $n \in \mathbb{N}$, we can also find sufficiently large $m \in \mathbb{N}$ such that $n > m$ and

$$\{1, \ldots, m\} \subset \{f(1), \ldots, f(n)\}.$$

Then

$$|t_n - L| \leq |t_n - s_m| + |s_m - L| \leq \sum_{k=m+1}^{\infty} |x_k| + |s_m - L|$$

and so the last two terms tend to zero. So the term also tends to zero and gives $\lim\ t_n = L$.

□

Exercise 3.8. Let $\sum\ x_n$ be an absolutely convergent series. Let $\sum\ y_n$ be a rearrangement of $\sum\ x_n$. Deduce that $\sum\ y_n$ is also absolutely convergent by using Theorem 3.7.

3.3. **Conditional convergence.** We now discuss the converse of Theorem 3.3.

Definition 3.9. A series $\sum\ x_n$ is said to be conditionally convergent if $\sum\ x_n$ is convergent, but $\sum\ x_n$ is not absolutely convergent.

The typical example of conditional convergence is the following:

Example 3.10. The series $\sum \frac{(-1)^n}{n}$ is conditionally convergent. We showed before that $\sum \frac{1}{n}$ is not convergent, and so we only have to show that $\sum \frac{(-1)^n}{n}$ is convergent, which we shall use a more general result in Theorem 3.13.

Example 3.11. The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is absolutely convergent, and so is *not* conditionally convergent.

Exercise 3.12. Explain the relations between convergence, absolute convergence and conditional convergence.

Theorem 3.13. *(Alternating Series Test) Let (x_n) be a decreasing sequence of strictly positive numbers with $\lim_{n\to\infty} x_n = 0$. Then the alternating series $\sum(-1)^{n+1}x_n$ converges.*

Proof. Let

$$s_n = \sum_{k=1}^{n}(-1)^{k+1}x_k.$$

Step 1: Show (s_{2k}) is increasing Let s_n be the partial sums. Then

$$s_{2k} = (x_1 - x_2) + (x_3 - x_4) + \ldots + (x_{2k-1} - x_{2k}).$$

Since (x_n) is decreasing, $x_{2k-1} - x_{2k} \geq 0$. Then s_{2k} is an increasing subsequence.

Step 2: Show (s_{2k}) is bounded above We also have:

$$s_{2k} = x_1 - (x_2 - x_3) - \ldots - (x_{2k-2} - x_{2k-1}) - x_{2k} < x_1$$

Hence, s_{2k} is bounded above.

Step 3: Show convergences By the MCT, the subsequence (s_{2k}) converges.

Note that $s_{2k+1} = s_{2k} + x_{2k+1}$. Since $\lim_{k\to\infty} s_{2k}$ exists and $\lim\ x_k$ exists, $\lim_{k\to\infty} s_{2k+1}$ also exists and $\lim_{k\to\infty} s_{2k+1} = \lim_{k\to\infty} s_{2k}$. This implies that $\lim\ s_n$ exists by discussions in Example 1.2(2) of Chapter 2 (Part 1). □

Exercise 3.14. Determine if the following series are absolutely convergent or conditionally convergent:

(a) $0 < \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}+1}$

(b) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2+1}$

The following example shows that the rearrangement theorem does not hold for conditional convergence.

Exercise 3.15. (a) Show that $0 < \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} < 1$.

(b) Consider the following rearrangement:

$$\left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) + \dots$$

Show that the rearrangement of the harmonic series is not equal to the original series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$.

3.4. **Root and Ratio Tests.**

Theorem 3.16. *(Root Test) Let (x_n) be a sequence. Let $r = \lim \sqrt[n]{|x_n|}$ (assume the limit exists).*

(a) *If $r < 1$, then $\sum\ x_n$ is absolutely convergent.*
(b) *If $r > 1$, then $\sum\ x_n$ is divergent.*

Proof. (Sketch) (a) Let $r_0 \in (r, 1)$ and let $\epsilon = r_0 - r$. There exists $K \in \mathbb{N}$ such that for any $n > K$,

$$|x_n| < r_0^n.$$

Since $\sum_n r_0^n$ converges, the comparison test implies that $\sum |x_n|$ also converges. This implies that $\sum\ x_n$ converges.

(b) There exists $K \in \mathbb{N}$ such that for any $n > K$,

$$\sqrt[n]{|x_n|} > 1 \Rightarrow |x_n| > 1.$$

Hence, $\lim\ |x_n| \neq 0$. By the n-th test, the series $\sum\ x_n$ diverges. □

Theorem 3.17. *(Ratio Test) Let (x_n) be a sequence of non-zero numbers. Let $r = \lim \frac{|x_{n+1}|}{|x_n|}$ (assume the limit exists).*

(a) *If $r < 1$, then $\sum\ x_n$ is absolutely convergent.*
(b) *If $r > 1$, then $\sum\ x_n$ is divergent.*

The proof relies on a similar idea of the proof of previous theorem. For details, refer to Page 272 of the book of Bartle-Sherbert.

Example 3.18. Let $x_n = \frac{1}{(-3)^n}$ be the geometric sequence. Then

(1) $\sqrt[n]{|x_n|} = \frac{1}{3} < 1$ and so, by the root test, the geometric series $\sum\ x_n$ is absolutely convergent.
(2) $|\frac{x_{n+1}}{x_n}| < 1$ and so, by the ratio test, the geometric series $\sum\ x_n$ is absolutely convergent.

Exercise 3.19. Use Root/Ratio Tests to determine whether the following series is absolutely convergent or divergent:

(1) $\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n n}$;

(2) $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{n^2}$ (Hint: use root test).